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5th Sem / Mechanical Engg.
Sub : Refrigeration and Air Conditioning

Time : 3Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The ratio of heat extracted in the refrigerator to the work done on the refrigerator is called C.O.P. of ________. (CO1)

a) Heat pump b) Refrigerator
c) Heat engine d) None of these

Q.2 C.O.P. of domestic air conditioner as compared to that of domestic refrigerator is (CO1)

a) Lower b) Higher
c) Same d) Unpredictable

Q.3 The ratio of high temperature to low temperature for reversed carnot cycle refrigeration is 1.25. The C.O.P. will be (CO2)

a) 2 b) 3
c) 4 d) 5

Q.4 The refrigerant widely used in window air conditioners is (CO2)

a) R-12 b) R-22
c) R-717 d) R-744

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Q.5 During summer comfort cooling, the relative humidity of the air should not above (CO4)

a) 30% b) 40%

c) 50% d) 60%

Q.6 The difference between dry bulb temperature and wet bulb temperature is known as: (CO3)

a) Dew point depression
b) Dry bulb depression
c) Wet bulb depression
d) None of these

Q.7 Fluid used in Electrolux refrigerator is: (CO1)

a) Water, ammonia, Hydrogen
b) Ammonia Hydrogen
c) Water, Hydrogen
d) None of these

Q.8 The curved lines that follow the saturation curve along the toe of a psychrometric chart represent: (CO3)

a) Wet bulb temp, lines
b) Dry bulb temp. Lines
c) Relative Humidity lines
d) Dew Point Temp lines

Q.9 Cooling towers are used in (CO5)

a) Water cooled condenser
b) Air cooled condenser
c) Evaporator
d) None of these

Q.10 What is the maximum star rating air conditioner available in market. (CO5)

a) 2 star b) 3 star

c) 4 star d) 5 star

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 In throttling process, enthalpy remains____ (CO1)

Q.12 The sub-cooling of refrigerant increases the C.O.P. (True/False) (CO1)

Q.13 When the liquid evaporates, it gives cooling effect. (True/False) (CO2)

Q.14 Define Refrigerant. (CO2)

Q.15 The chemical name of R-12 is ______ (CO5)

Q.16 Define condenser. (CO3)

Q.17 Define dry bulb temperature. (CO3)

Q.18 Define refrigeration. (CO1)

Q.19 Write full form of C.O.P. (CO4)

Q.20 Define humidification. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Write short note on natural and artificial refrigeration. (CO1)

Q.22 Explain use of flash chamber in vapour compression refrigeration cycle. (CO1)

Q.23 Draw p-v diagram of simple vapour compression refrigeration cycle. (CO2)

Q.24 Write the properties of R-22. (CO2)

Q.25 Write short note on domestic Electrolux refrigeration system. (CO5)

Q.26 Write short note on overload protector. (CO4)

Q.27 Explain the working of Thermostatic expansion valve. (CO5)

Q.28 Explain use of compressor in refrigeration systems. (CO5)

Q.29 Explain the function of expansion valve in refrigeration system. (CO5)

Q.30 Define psychrometry chart. Show specific humidity lines and relative humidity lines on psychrometry chart. (CO3)

Q.31 Explain process of cooling with dehumidification. (Co3)

Q.32 Explain sensible heat factor. (CO4)

Q.33 Define sensible heat and latent heat. (CO5)

Q.34 Write short note on auto-defrosting. (CO5)

Q.35 A machine working on Carnot cycle operates between 305K and 260K. (CO4)

Determine the C.O.P. when it is operated as i) a refrigerating machine ii) A heat pump iii) A heat Engine

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Write the advantages and disadvantages of air refrigeration system over vapour compression system. (CO1)

Q.37 Explain simple vapour absorption system with the help of neat sketch. (CO2)

Q.38 Explain the working principle of reciprocating compressor with neat sketch. (CO5)